Biliary Atresia, Part Ten: Hepatopulmonary Syndrome and Portopulmonary Hypertension. Screening with Oximetry and Echocardiography, the Splenic Malformation, and the Timing of Transplantation.

Imagine a nine-year-old boy who had a Kasai operation for biliary atresia at fifty-two days of age. His jaundice cleared, and for years he has been one of the clinic's success stories. He was born with polysplenia, several small spleens instead of one. His bilirubin is normal, his albumin is normal and his clotting is normal. But his mother says he now stops halfway up the stairs at school, and when you look at his hands you see that his fingertips have become clubbed. His oxygen saturation is ninety-four percent sitting and falls to ninety percent when he stands up. His platelet count is ninety thousand per microlitre. The question in front of you is whether this is his liver, and if so, what it means for him.

The main idea of this chapter is that the lungs can become the most important organ in a child with biliary atresia even when the liver seems to be doing well. Two different lung vascular disorders arise from liver disease and portal hypertension. The first is hepatopulmonary syndrome, in which the small blood vessels of the lung widen so much that blood passes through without picking up enough oxygen. The second is portopulmonary hypertension, in which the pulmonary arteries narrow and scar, so that the pressure in the lungs rises and the right side of the heart begins to fail. Both grow out of portal hypertension and the shunting of blood around the liver, not out of liver failure. That is why they appear in the children who look best, the jaundice-free survivors with their native liver. Hepatopulmonary syndrome is cured by liver transplantation and is a reason to transplant. Portopulmonary hypertension is not reliably cured by transplantation, and when it is severe it makes transplantation too dangerous to attempt. Screening for both is cheap, and missing either can be fatal.

A few terms first. Portal hypertension means raised pressure in the portal vein, the vessel that carries blood from the intestines and the spleen to the liver. Portosystemic shunting means that some of this blood takes detours through collateral veins and reaches the general circulation, and therefore the lungs, without passing through the liver. Pulse oximetry is the clip on the finger that measures oxygen saturation. An arterial blood gas measures the partial pressure of oxygen in arterial blood directly, and from it one can calculate the alveolar to arterial oxygen gradient, which is the difference between the oxygen in the air sacs and the oxygen in the blood; a large gradient means oxygen is not crossing properly. Contrast echocardiography, sometimes called a bubble study, is an ultrasound of the heart during

which agitated saline containing tiny bubbles is injected into a vein. Right-heart catheterisation is the direct measurement of pressures in the pulmonary artery through a catheter, and it is the only way to diagnose pulmonary hypertension with certainty.
The evidence in this chapter comes from several sources. The definitions come from the European Respiratory Society task force on pulmonary-hepatic vascular disorders, published in 2004, from the International Liver Transplantation Society practice guidelines on both syndromes published in 2016, and from the society's 2025 update on portopulmonary hypertension, which aligned its criteria with the 2022 European Society of Cardiology and European Respiratory Society guidelines on pulmonary hypertension. Data on children with biliary atresia come from the Japanese Biliary Atresia Registry, analysed by Okubo and Nio in 2025, from a cohort in Osaka reported by Ueno in 2017, from Seoul reported by Kim in 2021, from Toronto reported by Noli in 2008, from Essen reported by Hoerning in 2014, from Birmingham reported by Warner in 2018, and from Denver, where Condino and Ivy described children with portopulmonary hypertension in 2005. Transplant outcomes come from an analysis of the United States Scientific Registry of Transplant Recipients by Raza and colleagues in 2023. And the paediatric transplant perspective comes from the 2026 practice guideline on evaluating children for liver transplantation from the American Association for the Study of Liver Diseases, the American Society of Transplantation and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition.

Part one. One portal circulation, two lung diseases

Why should a liver problem harm the lungs? The healthy liver acts as a filter. Blood from the gut carries many substances, including bacterial products and signalling molecules, that the liver normally clears before they reach the rest of the body. When portal hypertension forces blood through collaterals around the liver, those substances reach the lungs unfiltered. At the same time, the diseased liver itself releases more of certain mediators.
The best described pathway for hepatopulmonary syndrome runs as follows. The liver releases more endothelin-one, a signalling peptide. In the lung, endothelin-one acts on a receptor called the endothelin B receptor on the inner lining of blood vessels, and this increases the production of nitric oxide, a powerful vasodilator. Macrophages that collect inside the lung vessels add further vasodilators, including carbon monoxide made by the enzyme haem oxygenase-one, and they release vascular endothelial growth factor, which drives the growth of new vessels. The result is that the smallest arteries and capillaries of the lung widen dramatically.
It helps to picture what that does to oxygen. In a normal capillary, a red cell passes so close to the air sac that oxygen diffuses into it almost at once. In a widened capillary, red cells in the centre of the stream are too far from the air for oxygen to reach them in the time available. Some vessels widen so much that they behave like true shunts, with blood passing

straight from artery to vein. Either way, blood leaves the lung less than fully oxygenated, and the child becomes hypoxaemic. These widened vessels are concentrated at the bases of the lungs, which receive more blood when a person is upright. That explains a characteristic sign called orthodeoxia: the oxygen saturation falls when the child sits or stands and improves when the child lies down.

Portopulmonary hypertension starts from the same portal circulation but goes in the opposite direction. Here the pulmonary arteries constrict, their walls thicken and proliferate, and eventually they develop the same plexiform lesions seen in other forms of pulmonary arterial hypertension. The resistance to flow through the lungs rises, the pulmonary artery pressure climbs, and the right ventricle, which is built to pump against low pressure, enlarges and eventually fails. The child may present with breathlessness, chest pain, fainting on exertion, or sudden death.

The key clinical lesson from this biology is that the driver is the shunt, not the loss of liver function. Hepatopulmonary syndrome occurs in children with normal albumin and normal clotting, in children with portal hypertension but no cirrhosis, and in children with congenital portosystemic shunts, where closing the abnormal vessel can reverse it. In the Seoul cohort, the standard liver severity scores, the paediatric end-stage liver disease score and the model for end-stage liver disease, did not predict which children had hepatopulmonary syndrome. So the child at risk is exactly the one with a successful Kasai who nevertheless has significant portal hypertension.

Part two. Definitions

Hepatopulmonary syndrome requires three things. First, liver disease or portal hypertension. Second, evidence of widened lung vessels, which in practice means a positive bubble study in which bubbles appear in the left side of the heart three or more heartbeats after they first appear in the right side. The delay matters: bubbles that cross within one or two beats suggest a hole in the heart, such as a patent foramen ovale, which is common in infants, whereas late bubbles have travelled through the lungs. Third, a gas exchange defect, defined by the 2016 guidelines as an alveolar to arterial oxygen gradient of fifteen millimetres of mercury or more while breathing room air and sitting. A nuclear medicine scan using technetium-labelled macroaggregated albumin can support the diagnosis. These albumin particles are normally trapped in the lung capillaries; if more than six percent of them reach the brain, they must have passed through widened lung vessels. This scan is especially useful when the child also has lung disease. Severity is graded by the arterial oxygen pressure: eighty millimetres of mercury or more is mild, sixty to seventy-nine is moderate, fifty to fifty-nine is severe, and below fifty is very severe.

Portopulmonary hypertension is defined by right-heart catheterisation in a patient with portal hypertension. The 2016 guideline used a mean pulmonary artery pressure above

twenty-five millimetres of mercury and a pulmonary vascular resistance above three Wood units, with a normal pulmonary artery wedge pressure, which shows that the problem lies in the lung arteries rather than in the left heart. The 2025 update lowered the thresholds to a mean pressure above twenty and a resistance above two Wood units, with a wedge pressure of fifteen or less. Severity bands are now mild at twenty to thirty-four, moderate at thirty-five to forty-four, and severe at forty-five or above.

There is no definition specific to children. Arterial blood gases are often avoided in children, so paediatric studies mix pulse oximetry, capillary blood gases, bubble studies and nuclear scans. When you compare rates between papers, check how each one defined the syndrome. And remember that a positive bubble study with normal oxygenation is intrapulmonary vascular dilatation, a precursor or marker, but not yet hepatopulmonary syndrome. The two syndromes can also coexist or follow one another in the same child.

Part three. How common, and when

In unselected populations these syndromes are rare. In the Japanese registry of three thousand seven hundred and seventy-seven children with biliary atresia, hepatopulmonary syndrome was recorded in one point nine percent, which is seventy-one children, and portopulmonary hypertension in zero point seven percent, which is twenty-seven children. In Osaka, among eighty-eight children with biliary atresia, the rates were six point eight percent and three point four percent. In Seoul, eight of one hundred and twenty children with biliary atresia, about seven percent, had hepatopulmonary syndrome.

The rate depends heavily on who is screened and how. In Toronto, a prospective study of three hundred and five children with liver disease found hepatopulmonary syndrome in only two, under one percent, but in two of the twenty-six children with cirrhosis or portal hypertension who had a bubble study, about eight percent. Seventeen percent of the Toronto children who had bubble studies showed widened lung vessels despite normal blood gases. And in Essen, among forty-five children with cirrhosis, forty percent had intrapulmonary vascular dilatation. So the honest summary is this: in registries the syndromes are rare, but when children with portal hypertension are actively screened with bubble studies, vascular dilatation is common and the full syndrome appears in several percent.

Timing is the second key fact. These are complications of long survival with the native liver. In Osaka, hepatopulmonary syndrome began at a median age of four years, and portopulmonary hypertension at a median of fifteen years. In Birmingham, among twenty children with hepatopulmonary syndrome, nine of whom had biliary atresia, the median age at diagnosis was ten years. In Seoul, children with biliary atresia and hepatopulmonary syndrome were transplanted at a median age of thirteen years, against one year for the other children with biliary atresia. In Denver, the median age at diagnosis of portopulmonary hypertension was about thirteen and a half years, roughly twelve years after portal

hypertension had first been recognised. The Japanese registry shows the same pattern: hepatopulmonary syndrome begins to appear around the second year of follow-up and peaks around the fifth, while portopulmonary hypertension appears around the fifth year and peaks around the fifteenth. In short, hepatopulmonary syndrome is a disease of childhood and portopulmonary hypertension a disease of adolescence. A clinic that stops looking after the first few years will meet portopulmonary hypertension for the first time when a teenager faints on the sports field or collapses at the induction of anaesthesia.

Part four. The splenic malformation

About one in ten children with biliary atresia in Europe and North America, and fewer than three in a hundred in East Asia, have the syndromic form known as biliary atresia splenic malformation. As defined by the King's College group in 1993, it combines polysplenia or absence of the spleen with other anomalies such as reversal of the abdominal organs, a portal vein that runs in front of the duodenum, or absence of the part of the inferior vena cava that normally passes behind the liver.

This group appears to carry a higher risk of hepatopulmonary syndrome. In Seoul, polysplenia or an interrupted inferior vena cava was present in three of the eight children with hepatopulmonary syndrome but in only one of the hundred and twelve children without it, and it was the only independent risk factor, with a p value of zero point zero zero one. A series from Riyadh, published in 2010, also found polysplenia over-represented among children with the syndrome.

Why might this be? When the inferior vena cava is interrupted, blood from the lower body returns through the azygos vein, and the hepatic veins may drain directly into the heart. Cardiologists have long known that diverting hepatic venous blood away from one lung, as happens after certain heart operations such as the Glenn procedure, causes arteriovenous malformations in that lung. The explanation offered is that hepatic venous blood carries a protective factor that the lung needs. Abnormal venous anatomy in the splenic malformation may distribute that factor unevenly. Congenital portosystemic shunts are also more common in this group and are a well-known cause of hepatopulmonary syndrome. The practical message is to have a lower threshold for bubble studies in children with the splenic malformation, to look for a portosystemic shunt on Doppler ultrasound or cross-sectional imaging, and to map the venous anatomy early because it will matter for any future transplant.

Part five. Screening with oximetry, and its limits

Pulse oximetry is the recommended first screen. The 2016 guideline suggests that a saturation below ninety-six percent at rest is a reasonable trigger for an arterial blood gas. The 2026 paediatric transplant evaluation guideline recommends screening every child

evaluated for transplantation with pulse oximetry, measured both lying down and upright, to detect orthodeoxia. A fall of four percentage points or more on standing is a strong clue. The weakness of oximetry in children is that it detects established disease, not early disease. In Essen, every child who had intrapulmonary vascular dilatation on bubble study had a saturation above ninety-eight percent. Capillary blood gases did better, with a sensitivity of ninety-four percent but a specificity of only fifty-three percent for a raised alveolar to arterial gradient. Two children with moderate hepatopulmonary syndrome in that cohort died before transplantation. In Denver, a study of thirty-three children with portal hypertension and no symptoms, nineteen of them with biliary atresia, living at high altitude, found intrapulmonary vascular dilatation in nineteen percent, hepatopulmonary syndrome in one child and portopulmonary hypertension in none; the authors recommended oximetry at every visit with targeted bubble studies. And oximetry tells you nothing about portopulmonary hypertension, which can be severe with a perfectly normal saturation.
A practical diagnostic pathway therefore runs like this. Suspect the syndrome in any child with clubbing, cyanosis, breathlessness, breathlessness that is worse upright, a saturation below ninety-six percent, or a fall on standing, and in every transplant candidate. Measure gas exchange with an arterial gas on room air while sitting, accepting a capillary gas as a sensitive but less specific substitute. Perform a bubble study and look for late bubbles. Use the macroaggregated albumin scan to confirm and quantify the shunt, especially when there is lung disease, and to provide a baseline. Grade severity by the arterial oxygen pressure. And look for a congenital portosystemic shunt that could be closed. Remember the pitfalls: early bubbles suggest a hole in the heart; anaemia, pneumonia and lung compression from ascites can lower the saturation for other reasons; and a positive bubble study with normal gases needs follow-up, not listing.

Part six. Screening for portopulmonary hypertension

For portopulmonary hypertension the screening test is transthoracic echocardiography, which estimates the right ventricular systolic pressure from the speed of blood leaking back through the tricuspid valve. The 2016 guideline recommended right-heart catheterisation when the estimated pressure exceeded fifty millimetres of mercury. The 2025 update lowered that trigger: an estimated right ventricular systolic pressure above forty millimetres of mercury, a tricuspid regurgitation velocity above two point eight metres per second, or more than mild right ventricular enlargement or dysfunction should now prompt referral to a pulmonary hypertension team and catheterisation. The update recommends screening all candidates for liver transplantation and for a transjugular intrahepatic portosystemic shunt, and says yearly echocardiography on the waiting list is reasonable. It contains no section on children.

In small children the echocardiographic estimate is less reliable, so the study should be read by a paediatric cardiologist, who will also look at flattening of the septum between the ventricles, the size of the right ventricle and the time it takes blood to accelerate in the pulmonary artery. The message is simple: a raised pressure estimate in a child with portal hypertension is a question for a catheter, not for watchful waiting, because survival depends on finding the disease before the mean pressure reaches forty-five.

Part seven. Hepatopulmonary syndrome and transplantation

There is no proven medical treatment for hepatopulmonary syndrome. Supplemental oxygen keeps the saturation above eighty-eight percent and supports exercise and growth, but the cure is liver transplantation. In the United States, a patient whose arterial oxygen pressure is below sixty millimetres of mercury on room air, with a shunt demonstrated and no significant lung disease, receives standardised exception points on the waiting list, because the laboratory score would otherwise never reflect the danger.

The registry analysis by Raza and colleagues looked at one hundred and twenty-four children transplanted with such an exception, at a median age of nine years. Nearly eighty-eight percent of them had a laboratory score below fifteen, so for most of them the exception was the only route to a graft. Their waiting list mortality was the same as that of other children. Five-year patient survival was eighty-eight point seven percent, compared with ninety-three point seven percent for children without the exception, an adjusted hazard ratio of one point seven five. The most striking finding was that very severe hypoxaemia did not worsen survival in children. Five-year survival was eighty-nine point four percent when the oxygen pressure was below fifty, ninety-two point one percent when it was fifty to fifty-nine, and seventy-six point five percent when it was sixty to sixty-nine, differences that were not statistically significant. That contrasts with adult data, in which very severe hypoxaemia predicts a worse outcome. In children, severity is a reason to list promptly, not a reason to decline.

After transplantation the syndrome resolves. In Birmingham, it resolved in all fifteen transplanted children, with no recurrence and a survival of eighty-seven and a half percent over five to twenty years. In Seoul, the brain uptake on the albumin scan fell below six percent at a median of about seven and a half months. But the first weeks can be difficult. Post-transplant hypoxaemia is the main early danger, oxygen may be needed for months, and in Birmingham a haemoglobin of one hundred and thirty grams per litre or more was associated with a longer intensive care stay and longer oxygen need. The anaesthetist and intensivist should know in advance, because severe post-transplant hypoxaemia may require inhaled nitric oxide, head-down positioning, prone ventilation or extracorporeal membrane oxygenation. Rarely, hepatopulmonary syndrome has appeared for the first time after living-donor transplantation, so problems with the graft's venous outflow or persistent

shunts should be considered if hypoxaemia returns.

Part eight. Treating portopulmonary hypertension

Portopulmonary hypertension is managed with a pulmonary hypertension team, following the 2025 update, which is based on adult data. The first step is to stop non-selective beta-blockers such as propranolol, because by lowering cardiac output they worsen the haemodynamics, and to manage varices with band ligation instead. A transjugular intrahepatic portosystemic shunt should be avoided, because it increases the volume of blood returning to an already struggling right heart. When the mean pulmonary artery pressure is below forty-five, monotherapy is used, with either a phosphodiesterase-five inhibitor such as sildenafil, preferred in advanced liver disease, or an endothelin receptor antagonist. When the mean pressure is forty-five or more, combination therapy is started from the outset, including a prostacyclin analogue when disease is severe or the right ventricle is failing. The only randomised trial in this condition, called PORTICO and published in 2019, gave macitentan to eighty-five adults and reduced pulmonary vascular resistance by thirty-five percent at twelve weeks, without changing the six-minute walk distance. Bosentan is the endothelin antagonist most associated with liver injury and needs regular liver tests. The newer drug sotatercept has not been studied in this condition.
Treatment determines transplant candidacy. A mean pressure above forty-five to fifty, a resistance above five Wood units, or severe right ventricular dysfunction is an absolute contraindication to liver transplantation. In the United States, a child can receive exception points if the baseline mean pressure is at least thirty-five and the resistance above three Wood units, and if treatment then brings the mean pressure below thirty-five with a resistance below five, or between thirty-five and forty-five with a resistance below three. Catheterisation is repeated every three months while the exception is held. After transplantation, pulmonary hypertension drugs are weaned no earlier than three to six months, and only about half of adults stop them completely.
The paediatric outcome data show what happens when the disease is found late or early. In Denver in 2005, four of seven children died, one of them at the induction of anaesthesia for transplantation. In Osaka, two of three children with biliary atresia and portopulmonary hypertension died, whereas all the children with hepatopulmonary syndrome survived. In contrast, a modern series of six children from Geneva, published in 2021, with a median mean pulmonary pressure of forty-seven and a resistance of six point six Wood units at diagnosis, treated with upfront combination therapy followed by transplantation or closure of a shunt, had no deaths. Portopulmonary hypertension is treatable when it is looked for.

Part nine. When the lungs set the clock

In biliary atresia these syndromes are often the only transplant indication in a child with near-normal bilirubin, albumin and clotting. For hepatopulmonary syndrome, once the arterial oxygen pressure is below sixty, list the child with exception points, and do not wait for further decline: the graft cures the lungs, children with very low oxygen still do well, and delay costs growth and exercise capacity. For portopulmonary hypertension, treat first, then transplant inside the window in which the pressures have been brought under control; if the mean pressure stays above forty-five to fifty despite treatment, transplantation is contraindicated. For children with the splenic malformation, map the inferior vena cava, the hepatic veins and the portal vein early, since a living-donor operation takes time to plan. The Japanese registry illustrates the consequence of all this: almost no hepatopulmonary syndrome or portopulmonary hypertension remains among twenty-year native-liver survivors, because those children were transplanted before the age of twenty.

How do the guidelines compare? The transplant society guidelines of 2016 and 2025, the 2016 European Association for the Study of the Liver guideline on vascular diseases of the liver, and the 2026 North American paediatric transplant evaluation guideline all agree that transplant candidates should be screened for hepatopulmonary syndrome. They differ in how: the adult documents describe oximetry followed by blood gas and bubble study, while the paediatric guideline names pulse oximetry, lying and upright. They differ on the echocardiographic trigger for catheterisation, which the 2025 update has lowered to forty millimetres of mercury. They align on stopping beta-blockers and avoiding a transjugular shunt in portopulmonary hypertension. There is a gap on surveillance of native-liver survivors, which none of the documents addresses, and a material difference in that every threshold applied to children is borrowed from adults. Neither the European nor the North American paediatric gastroenterology societies have published a specific statement on these syndromes.

Part ten. Traps, numbers and decisions

Several traps recur. The first is reassurance from a good liver: a jaundice-free child with a low liver score can still have a dangerous shunt. The second is reassurance from a normal saturation, which misses early hepatopulmonary syndrome and all early portopulmonary hypertension. The third is labelling breathlessness and clubbing as asthma or deconditioning; clubbing is never asthma. The fourth is continuing propranolol in a child with portopulmonary hypertension. The fifth is taking a child to theatre on echocardiographic estimates alone, without a catheter. And the biggest is treating these syndromes as pulmonary curiosities rather than as transplant decisions.

To bring the numbers together: in the Japanese registry, hepatopulmonary syndrome affects about two percent and portopulmonary hypertension under one percent of children with biliary atresia, while cohorts that screen find hepatopulmonary syndrome in about seven

percent. Hepatopulmonary syndrome begins at a median of four years and portopulmonary hypertension at a median of fifteen. In Seoul, polysplenia or an interrupted vena cava was present in three of eight children with hepatopulmonary syndrome against one of a hundred and twelve without. The diagnosis needs late bubbles and a gradient of fifteen millimetres of mercury or more, supported by more than six percent brain uptake on the albumin scan. In Essen every child with lung vessel dilatation had a saturation above ninety-eight percent. Portopulmonary hypertension is now defined by a mean pressure above twenty, a wedge pressure of fifteen or less and a resistance above two Wood units, and an echocardiographic pressure above forty should trigger a catheter. Children transplanted for hepatopulmonary syndrome had a five-year survival of nearly eighty-nine percent, and an oxygen pressure below fifty did not worsen it. The syndrome resolved in all fifteen transplanted children in Birmingham. And in portopulmonary hypertension, stop beta-blockers, avoid a transjugular shunt and treat, remembering that a mean pressure of forty-five to fifty or a resistance above five forbids transplantation.

Consider five common situations. For the nine-year-old boy at the start of this chapter, with polysplenia, clubbing and a saturation that falls on standing, arrange a blood gas, a bubble study and an albumin scan, and refer for transplantation if his oxygen pressure is below sixty. For a six-year-old with a good Kasai, a large spleen and a saturation of ninety-nine percent, arrange a yearly bubble study with an estimate of right heart pressure, because oximetry misses early disease. For a fourteen-year-old who faints during sport and whose echocardiogram estimates a right ventricular pressure of fifty-five, stop the propranolol, arrange a right-heart catheter and involve the pulmonary hypertension team, because this is probably portopulmonary hypertension and missing it can be fatal at anaesthesia. For a child with portopulmonary hypertension whose mean pressure is thirty-eight with a resistance of four Wood units on combination therapy, continue treatment, repeat the catheter and list when the criteria are met. And for a child with bubbles on the echocardiogram but an oxygen pressure of ninety-two, call it intrapulmonary vascular dilatation, not hepatopulmonary syndrome, and follow with oximetry and gases.

What should you do differently in clinic? Record the oxygen saturation, sitting and standing, at every visit and write both values in the letter, because a slow drift from ninety-eight to ninety-five percent over two years is the earliest visible sign. Look at the fingers for clubbing and ask about breathlessness, chest pain and fainting. In every native-liver survivor with established portal hypertension, arrange a yearly bubble study with an estimate of right heart pressure, and continue it through adolescence. Have a lower threshold in children with the splenic malformation or a portosystemic shunt. Stop beta-blockers if pulmonary hypertension is suspected. And involve the transplant centre at the moment either syndrome is diagnosed, because in biliary atresia the lungs, not the liver, may decide when a child needs a new liver.